

Duality I: Bounds and the Dual Problem

A feasible solution tells us what we can achieve. A dual solution tells us what no feasible solution can beat.

Our route: combine constraints $\rightarrow$ choose the best weights $\rightarrow$ write the dual $\rightarrow$ prove weak duality.

Minimization and its lower bounds · Bertsimas and Tsitsiklis §§4.1–4.3.

Start with arithmetic

Nutrition: prove the minimum cost

Compare the forms and explain the signs

Weak duality

Optional derivations

A good solution and a good bound answer different questions

For a minimization problem, a feasible point x gives an **upper bound** on the optimal value v^* .

$$v^* \leq c^\top x.$$

A **lower bound** L proves that every feasible solution costs at least L .

$$L \leq v^* \leq c^\top x.$$

For example, if the objective is $x_1 + 3x_2$ and a constraint says $x_1 + 3x_2 \geq 2$, the lower bound $L = 2$ is immediate.

When no constraint directly matches the objective, combine several constraints.

Start by adding inequalities

Primal minimization: $\geq$ constraints; variables unrestricted.

$$\min x_1 + 3x_2 \quad \text{s.t.} \quad x_1 + x_2 \geq 2, \quad x_2 \geq 1.$$

Unrestricted means no separate variable-sign restrictions; the listed constraints still apply.

Multiply the second inequality by 2, then add:

$$(x_1 + x_2) + 2x_2 \geq 2 + 2 \cdot 1.$$

$$x_1 + 3x_2 \geq 4.$$

Every feasible solution costs at least 4. We obtained a bound without running simplex.

The point $(1, 1)$ is feasible and has cost 4, so it attains this bound.

Another constraint can give a stronger bound

Primal minimization: $\geq$ constraints; variables unrestricted.

$$\begin{array}{ll}\min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2, \quad x_2 \geq 1, \quad x_1 - x_2 \geq 3.\end{array}$$

Here x_1, x_2 are unrestricted variables: no additional sign restrictions are imposed.

$$0(x_1 + x_2) + 4x_2 + (x_1 - x_2) \geq 0 \cdot 2 + 4 \cdot 1 + 3.$$

$$x_1 + 3x_2 \geq 7.$$

Weights $(0, 4, 1)$ improve the previous bound 4 to 7. The feasible point $(4, 1)$ attains 7.

Which constraint weights give a valid bound?

Primal minimization: $\geq$ constraints; variables unrestricted.

Matching coefficients gives $(p_1, p_2, p_3) = (t, 4 - 2t, 1 - t)$.

t	Weights	Weighted RHS	Certificate?
1	$(1, 2, 0)$	4	Yes
0	$(0, 4, 1)$	7	Yes
$-1/2$	$(-1/2, 5, 3/2)$	$17/2$	No: a negative weight

Valid weights have $0 \leq t \leq 1$, so $7 - 3t \leq 7$. The best bound is 7, attained at $t = 0$.

Multiplying a $\geq$ inequality by a negative number reverses its direction; such a term cannot be added as another $\geq$ inequality.

The best bound is itself a linear program

Primal minimization: $\geq$ constraints; variables unrestricted.

Weight signs: the $\geq$ constraints need nonnegative weights to keep the weighted sum $\geq$ its right-hand side.

Coefficient conditions: unrestricted variables require exact matching to objective coefficients 1, 3.

$$\begin{array}{ll}\max & 2p_1 + p_2 + 3p_3 \\ \text{s.t.} & p_1 + p_3 = 1, \\ & p_1 + p_2 - p_3 = 3, \\ & p_1, p_2, p_3 \geq 0.\end{array}$$

The original problem is the **primal**. This problem of finding the strongest weighted bound is its **dual**.

Can we meet the requirements for less than \$14?

Primal minimization: $\geq$ constraints; servings nonnegative.

Nutrient requirements are still minimums; now servings are nonnegative. Data are illustrative.

Per serving	Protein units	Carb units	Cost
Lentil dish x_L	2	1	\$4
Grain dish x_G	1	2	\$5
Snack x_S	1	1	\$4
Minimum required	4	5	

Minimize $4x_L + 5x_G + 4x_S$ while meeting both minimums. Fractional servings are allowed.

One feasible meal: 1 lentil dish + 2 grain dishes + no snack.
Protein: $2 + 2 = 4$. Carbs: $1 + 4 = 5$. Cost: $4 + 10 = 14$.

We know $v^* \leq 14$. Can any other combination cost less?

Combine the nutritional requirements

Primal minimization: $\geq$ constraints; servings nonnegative.

$$\begin{aligned} 2x_L + x_G + x_S &\geq 4 && \text{(protein),} \\ x_L + 2x_G + x_S &\geq 5 && \text{(carbohydrates).} \end{aligned}$$

Try this combination: the protein constraint plus twice the carbohydrate constraint.

$$(2x_L + x_G + x_S) + 2(x_L + 2x_G + x_S) \geq 4 + 2(5).$$

Since $x_S \geq 0$, the actual meal cost satisfies

$$\underbrace{4x_L + 5x_G + 4x_S}_{\text{meal cost}} \geq 4x_L + 5x_G + 3x_S \geq 14.$$

Our feasible meal costs \$14, so it is optimal.

Find the strongest cost guarantee

Primal minimization: $\geq$ constraints; servings nonnegative.

Weight signs: the protein and carbohydrate $\geq$ constraints give $p, q \geq 0$.

$$(2p + q)x_L + (p + 2q)x_G + (p + q)x_S \geq 4p + 5q.$$

Coefficient conditions: choose coefficients $\leq$ food costs. Multiplication by nonnegative servings preserves these inequalities.

max	$4p + 5q$	lower bound on meal cost
s.t.	$2p + q \leq 4$	lentil dish,
	$p + 2q \leq 5$	grain dish,
	$p + q \leq 4$	snack,
	$p, q \geq 0.$	

Weights $(1, 1)$ give a \$9 bound; $(1, 2)$ give \$14, matching our meal.

Our continuing example has equality constraints

Primal minimization: equality constraints; variables nonnegative.

$$\begin{array}{ll}\min & 13x_1 + 10x_2 + 6x_3 \\ \text{s.t.} & 5x_1 + x_2 + 3x_3 = 8, \\ & 3x_1 + x_2 = 3, \quad x_1, x_2, x_3 \geq 0.\end{array}$$

What changed? The variables remain nonnegative. The constraints are now equalities, so their weights may have either sign.

The coefficient conditions will still be $\leq$, because the primal variables remain nonnegative.

Two weighted equalities nearly reproduce the objective

Primal minimization: equality constraints; variables nonnegative.

Our equalities are $5x_1 + x_2 + 3x_3 = 8$ and $3x_1 + x_2 = 3$.

$$2(5x_1 + x_2 + 3x_3) + (3x_1 + x_2) = 19.$$

$$13x_1 + 10x_2 + 6x_3 = 19 + 7x_2 \geq 19.$$

The difference is $7x_2$, which is nonnegative. The weighted equalities give a lower bound even though the x_2 coefficient is not matched exactly.

Choose weights so the remaining terms are nonnegative

Primal minimization: equality constraints; variables nonnegative.

Multiply the equalities by unrestricted weights p_1, p_2 :

$$(5p_1 + 3p_2)x_1 + (p_1 + p_2)x_2 + 3p_1x_3 = 8p_1 + 3p_2.$$

Because $x_1, x_2, x_3 \geq 0$, a lower bound follows whenever

$$5p_1 + 3p_2 \leq 13, \quad p_1 + p_2 \leq 10, \quad 3p_1 \leq 6.$$

$$8p_1 + 3p_2 \leq 13x_1 + 10x_2 + 6x_3.$$

Each nonnegative primal variable supplies one coefficient inequality.

Maximize the bound subject to those coefficient tests

Primal minimization: equality constraints; variables nonnegative.

Weight signs: equalities allow unrestricted weights.

Coefficient conditions: nonnegative variables allow weighted coefficients $\leq$ objective coefficients.

$$\begin{array}{ll}\max & 8p_1 + 3p_2 \\ \text{s.t.} & 5p_1 + 3p_2 \leq 13, \\ & p_1 + p_2 \leq 10, \quad 3p_1 \leq 6, \\ & p_1, p_2 \text{ unrestricted.}\end{array}$$

Check $p = (2, 1)$: the three left sides are 13, 3, 6, all at most 13, 10, 6. Its bound is $8(2) + 3(1) = 19$.

Two primal equations give two dual variables. Three primal variables give three dual constraints.

Three examples, two separate decisions

All three: primal minimization → dual maximization.

Primal constraint → dual variable: choose the weight's sign.
Primal variable → dual constraint: compare its coefficients.

Example	Primal constraints	Primal variables	Dual variables	Dual constraints
Opening	$\geq$	Unrestricted	Nonnegative	$=$
Nutrition	$\geq$	Nonnegative	Nonnegative	$\leq$
Equality	$=$	Nonnegative	Unrestricted	$\leq$

Opening → nutrition: changing variable signs changes the coefficient conditions.

Nutrition → equality: changing constraint types changes the weight signs.

First: choose the sign of each constraint weight

For a minimization primal: after weighting, we want

weighted expression $\geq$ weighted right-hand side.

Original constraint	Allowed weight	Example after multiplication
$x \geq 2$	$p \geq 0$	$p = 2 : \quad 2x \geq 4$
$x \leq 2$	$p \leq 0$	$p = -1 : \quad -x \geq -2$
$x = 2$	p unrestricted	$p = -1 : \quad -x = -2$

Both weighted inequalities now point $\geq$, so we can add them. An equality also satisfies $\geq$. These are three separate illustrations.

Next: ensure that the weighted expression does not exceed the original objective.

Second: compare each variable’s coefficients

Still minimizing: the weighted expression must not exceed the objective.

For lentils, the weighted coefficient is $2p + q$, and the unit cost is 4:

$$2p + q \leq 4, \quad x_L \geq 0 \quad \implies \quad (2p + q)x_L \leq 4x_L.$$

Primal variable	Weighted coefficient vs. objective coefficient
$x_j \geq 0$	$\leq$: nonnegative multiplication preserves the comparison.
$x_j \leq 0$	$\geq$: nonpositive multiplication reverses the comparison.
x_j unrestricted	$=$: exact matching works for either sign.

weighted RHS $\leq$ weighted expression $\leq$ objective.

First comparison: constraint types and weight signs.

Second comparison: variable signs and coefficient conditions.

Write the same argument in matrix notation

Primal minimization: equality constraints; variables nonnegative.

$$\text{Primal: } \min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

Each equality supplies an unrestricted weight. The weighted coefficient of x_j is $p^\top A_j$, where A_j is column j of A .

$$\text{Dual: } \max b^\top p \quad \text{s.t.} \quad A^\top p \leq c, \quad p \text{ unrestricted.}$$

Nonnegative x_j requires $p^\top A_j \leq c_j$. Collecting these coefficient conditions gives $A^\top p \leq c$.

Practice: a primal with mixed signs

$$\begin{array}{ll}\min & x_1 + 2x_2 + 3x_3 \\ \text{s.t.} & -x_1 + 3x_2 = 5, \\ & 2x_1 - x_2 + 3x_3 \geq 6, \quad x_3 \leq 4, \\ & x_1 \geq 0, \quad x_2 \leq 0, \quad x_3 \text{ unrestricted.}\end{array}$$

- 1. Constraint types:** choose the sign of each weight.
- 2. Variable signs:** write one coefficient condition per variable.
- 3. Objective:** maximize the weighted right-hand side.

Use minimization rules. Constructing a dual does not assume feasibility or optimality.

Mixed signs: read one column at a time

Primal minimization → dual maximization. Read constraint types and variable signs separately.

Weight signs from the three constraints: equality → p_1 unrestricted; $\geq$ → $p_2 \geq 0$; $\leq$ → $p_3 \leq 0$.

Primal variable sign	Corresponding dual coefficient condition
$x_1 \geq 0$	$-p_1 + 2p_2 \leq 1$
$x_2 \leq 0$	$3p_1 - p_2 \geq 2$
x_3 unrestricted	$3p_2 + p_3 = 3$

$$\max \quad 5p_1 + 6p_2 + 4p_3.$$

Use the three dual constraints and the weight restrictions listed above.

If the primal maximizes, its dual gives an upper bound

This page changes convention: primal maximization $\rightarrow$ dual minimization.

$$\text{Primal: } \max c^\top x \quad \text{s.t.} \quad Ax \leq b, \quad x \geq 0.$$

$$\text{Dual: } \min b^\top p \quad \text{s.t.} \quad A^\top p \geq c, \quad p \geq 0.$$

$$c^\top x \leq p^\top Ax \leq p^\top b.$$

You can instead convert the objective to $\min(-c^\top x)$, then use the minimization rules. Track the sign of the final objective value.

From the next page onward, our primal is again a minimization problem.

Weak duality compares any feasible pair

Theorem 4.3 — Weak duality.

If x is primal feasible and p is dual feasible, then

$$b^\top p \leq c^\top x.$$

Neither solution needs to be optimal. Every dual feasible solution gives a lower bound on every primal feasible objective value.

Feasibility of both points is essential. A vector that violates a dual constraint does not provide this certificate.

The standard-form proof is one inequality

Assume $Ax = b$, $x \geq 0$, and $A^\top p \leq c$.

$$b^\top p = (Ax)^\top p = x^\top A^\top p.$$

Use primal feasibility, then regroup the same scalar product.

$$x^\top A^\top p \leq x^\top c = c^\top x.$$

Multiply each coefficient inequality by the corresponding nonnegative x_j , then add.

The proof compares the weighted constraints with the original objective. No simplex pivot is needed.

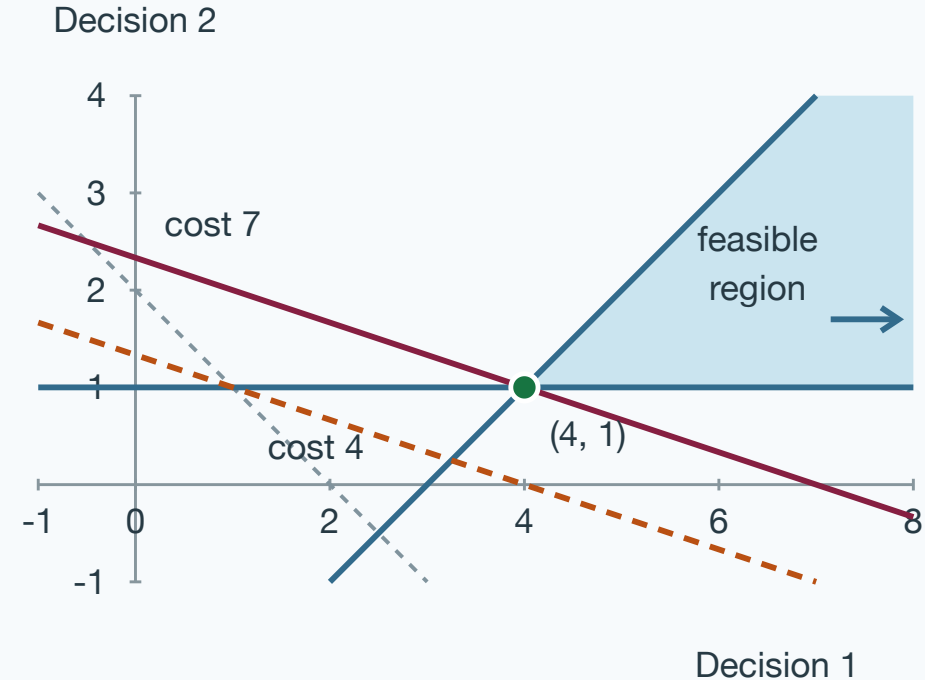
A lower bound lies below every feasible objective value

$$\begin{array}{ll}\min & x_1 + 3x_2 \\ \text{s.t.} & x_1 + x_2 \geq 2, \\ & x_2 \geq 1, \quad x_1 - x_2 \geq 3.\end{array}$$

Both variables are unrestricted. The plot returns to our opening example.

Weights $(1, 2, 0)$ give the lower bound 4. Its objective line does not touch the feasible region.

Weights $(0, 4, 1)$ give 7. This line touches the region at $(4, 1)$, which attains the bound.



Horizontal: x_1 ; vertical: x_2 . Blue boundaries: $x_2 = 1$ and $x_1 - x_2 = 3$. Gray: $x_1 + x_2 = 2$. The shaded set continues outside the window.

Matching feasible values certify optimality

Corollary 4.2. If a primal feasible point and a dual feasible point have the same objective value, both are optimal.

Call these feasible points $\hat{x}, \hat{p}$, with common value $c^\top \hat{x} = b^\top \hat{p} = v$.

Primal minimum. For every primal feasible x , weak duality gives

$$c^\top x \geq b^\top \hat{p} = v.$$

Our point $\hat{x}$ attains this lower bound, so it is optimal.

Dual maximum. For every dual feasible p , weak duality gives

$$b^\top p \leq c^\top \hat{x} = v.$$

Our point $\hat{p}$ attains this upper bound, so it is optimal.

A feasible meal costs \$14; dual weights prove no acceptable meal costs less. Both are optimal.

Unboundedness on one side rules out feasibility on the other

If the primal objective decreases without bound, the dual is infeasible. A finite dual lower bound would block that decrease.

If the dual objective increases without bound, the primal is infeasible. A finite primal value would block that increase.

These statements follow from weak duality. Their converses do not follow: infeasibility alone does not tell us that the other problem is unbounded.

A one-variable example makes the contradiction visible

Primal: $\min x_1 \quad \text{s.t.} \quad x_1 \leq 1, \quad x_1 \text{ unrestricted.}$

Feature of this minimization primal	Consequence for its dual
Constraint $x_1 \leq 1$	Nonpositive weight: $p_1 \leq 0$.
Unrestricted x_1 ; objective coefficient 1	Exact coefficient matching: $p_1 = 1$.
Right-hand side 1	Maximize $1 \cdot p_1$.

Dual: $\max p_1 \quad \text{s.t.} \quad p_1 \leq 0, \quad p_1 = 1.$

The dual conditions contradict each other. The primal is unbounded below: choose $x_1 = -M$ and let $M \rightarrow \infty$.

Practice: write the dual and try a certificate

$$\begin{array}{ll}\min & 2x_1 + x_2 \\ \text{s.t.} & x_1 + x_2 \geq 3, \\ & x_1 + 2x_2 \leq 6, \quad x_1 \geq 0, \quad x_2 \text{ unrestricted.}\end{array}$$

1. Write the dual, including every sign restriction.
2. Is $p = (1, 0)$ dual feasible? What bound does it give?
3. Check $x = (0, 3)$. What can you conclude?

Check feasibility before comparing objective values.

Practice solution: both sides attain 3

Constraint types: $\geq, \leq$ give $p_1 \geq 0, p_2 \leq 0$. **Variable signs:** $x_1 \geq 0$ gives $\leq$; unrestricted x_2 gives $=$.

$$\begin{array}{ll}\max & 3p_1 + 6p_2 \\ \text{s.t.} & p_1 + p_2 \leq 2, \quad p_1 + 2p_2 = 1, \\ & p_1 \geq 0, \quad p_2 \leq 0.\end{array}$$

For $p = (1, 0)$, the coefficient tests are $1 \leq 2$ and $1 = 1$; the lower bound is 3.

For $x = (0, 3)$, the primal rows give $3 \geq 3$ and $6 \leq 6$; the cost is 3. Weak duality therefore proves both points optimal.

Study guide: build, check, and interpret a dual

You should be able to:

- Construct a bound by weighting constraints.
- Explain each dual sign and coefficient inequality.
- Prove weak duality and check an optimality certificate.

Textbook: Bertsimas and Tsitsiklis, *Introduction to Linear Optimization*, §§4.1–4.3.

Optional derivations: the following appendix develops the Lagrangian viewpoint, dual-of-dual, equivalent formulations, and general weak duality.

Continue: Duality II asks why an optimal LP always has a matching dual certificate and how complementary slackness finds it.

Optional study: three ways to understand the same dual

Lagrangian: the dual chooses the best bound obtained after weighting the constraints.

Dual of the dual: applying the construction twice returns an equivalent primal problem.

Equivalent formulations: surplus variables, split free variables, and redundant rows change the notation while preserving the dual optimization problem.

The main lecture uses weighted arithmetic. These pages supply the longer derivations.

A Lagrangian packages the weighted-constraint argument

Start with $\min c^\top x$, subject to $Ax = b$, $x \geq 0$. Use unrestricted equality weights p and nonnegative sign-constraint weights s .

$$L(x, p, s) = c^\top x + p^\top (b - Ax) - s^\top x.$$

$$L(x, p, s) = p^\top b + (c - A^\top p - s)^\top x.$$

For a feasible x , the equality term vanishes and $-s^\top x \leq 0$. Hence $L(x, p, s) \leq c^\top x$.

A finite lower bound requires the remaining coefficient to vanish

Minimize the Lagrangian over **unrestricted** x . Here the sign constraints have already been included using $s \geq 0$.

$$g(p, s) = \inf_{x \in \mathbb{R}^n} L(x, p, s) = \begin{cases} p^\top b, & c - A^\top p - s = 0, \\ -\infty, & \text{otherwise.} \end{cases}$$

If a coefficient is nonzero, choose the sign of that x_j and make its magnitude arbitrarily large to drive L downward.

$$\max p^\top b \quad \text{s.t.} \quad A^\top p + s = c, \quad s \geq 0.$$

Eliminating s gives $A^\top p \leq c$, our standard-form dual.

For a maximization primal, construct an upper bound

Consider $\max c^\top x$, subject to $Ax \leq b$, $x \geq 0$. Choose $p \geq 0$.

$$L(x, p) = c^\top x + p^\top (b - Ax) = p^\top b + (c - A^\top p)^\top x.$$

For a primal feasible point, $L(x, p) \geq c^\top x$. Maximize L over $x \geq 0$:

$$\sup_{x \geq 0} L(x, p) = \begin{cases} p^\top b, & A^\top p \geq c, \\ +\infty, & \text{otherwise.} \end{cases}$$

The best finite upper bound is $\min b^\top p$ subject to $A^\top p \geq c$, $p \geq 0$.

The dual of the dual returns the primal

$$\text{P: } \min c^\top x \quad \text{s.t.} \quad Ax = b, \quad x \geq 0.$$

$$\text{D: } \max b^\top p \quad \text{s.t.} \quad A^\top p \leq c, \quad p \text{ unrestricted.}$$

Dualizing this maximization problem: its $\leq$ rows give $x \geq 0$; its unrestricted variables give $Ax = b$; its objective becomes $\min c^\top x$.

Theorem 4.1. The dual of the dual is equivalent to the original primal. Converting a maximization to minimization must include the objective-sign change.

Mixed-form example: convert its dual carefully

The mixed-form exercise had a maximization dual. Negate its objective and multiply its coefficient rows by -1 :

$$\begin{array}{ll}\min & -5p_1 - 6p_2 - 4p_3 \\ \text{s.t.} & p_1 - 2p_2 \geq -1, \\ & -3p_1 + p_2 \leq -2, \quad -3p_2 - p_3 = -3, \\ & p_1 \text{ unrestricted, } p_2 \geq 0, p_3 \leq 0.\end{array}$$

This minimization has the negative of the original dual objective value. Track that minus sign when interpreting the next dual.

Dualizing again recovers the original mixed problem

The dual of the minimization on the previous page is

$$\begin{array}{ll}\max & -x_1 - 2x_2 - 3x_3 \\ \text{s.t.} & x_1 - 3x_2 = -5, \\ & -2x_1 + x_2 - 3x_3 \leq -6, \quad -x_3 \geq -4, \\ & x_1 \geq 0, \quad x_2 \leq 0, \quad x_3 \text{ unrestricted.}\end{array}$$

Negating the objective to return to minimization, and reversing the displayed row signs, gives precisely the original mixed-form primal.

Its objective is $\min x_1 + 2x_2 + 3x_3$; its rows are $-x_1 + 3x_2 = 5$, $2x_1 - x_2 + 3x_3 \geq 6$, and $x_3 \leq 4$.

Adding a surplus variable preserves the dual

Start with $\min c^\top x$, subject to $Ax \geq b$, with x unrestricted. Its dual is $\max p^\top b$, subject to $A^\top p = c$, $p \geq 0$.

$$Ax - s = b, \quad s \geq 0, \quad \text{objective } c^\top x + 0^\top s.$$

Dualizing the equality formulation gives:

$$A^\top p = c, \quad -p \leq 0, \quad p \text{ initially unrestricted.}$$

The surplus columns enforce $p \geq 0$. We recover the same dual constraints and objective.

Splitting a free variable preserves its dual equality

Write an unrestricted vector as $x = x^+ - x^-$, with $x^+, x^- \geq 0$. The primal becomes

$$\min c^\top x^+ - c^\top x^- \quad \text{s.t.} \quad Ax^+ - Ax^- \geq b.$$

The two sets of nonnegative variables give two sets of dual coefficient inequalities:

$$A^\top p \leq c, \quad -A^\top p \leq -c, \quad p \geq 0.$$

Together they say $A^\top p = c$. The dual is unchanged.

A redundant equality only redistributes dual weights

Suppose row m is an exact combination of the first $m - 1$ rows, including its RHS:

$$a_m = \sum_{i=1}^{m-1} \gamma_i a_i, \quad b_m = \sum_{i=1}^{m-1} \gamma_i b_i.$$

For weights $p_1, \dots, p_m$, absorb the last weight into the others:

$$q_i = p_i + \gamma_i p_m \quad (i = 1, \dots, m - 1).$$

The weights are unrestricted because all primal rows are equalities. The next page checks both the bound and the coefficient tests.

The new weights give exactly the same bound and coefficients

$$p^\top b = \sum_{i=1}^{m-1} (p_i + \gamma_i p_m) b_i = \sum_{i=1}^{m-1} q_i b_i.$$

$$p^\top A = \sum_{i=1}^{m-1} (p_i + \gamma_i p_m) a_i^\top = \sum_{i=1}^{m-1} q_i a_i^\top.$$

Thus a feasible dual weight vector before removal gives one after removal with the same value.

Conversely, given q , choose $p_i = q_i$ for $i < m$ and $p_m = 0$. This recovers every feasible dual vector of the smaller system.

Equivalent formulations give equivalent duals

Theorem 4.2. The following transformations preserve the dual optimization problem up to equivalent variables and constraints:

1. Replace a free variable by the difference of two nonnegative variables.
2. Introduce a nonnegative slack or surplus to turn an inequality into an equality.
3. Remove a redundant equality from a feasible standard-form system.

The preceding pages provide the explicit correspondence in each case. The dual may look different on paper while describing the same attainable bounds.

Weak duality also holds with mixed row and variable signs

Let x, p satisfy the minimization primal and its dual. For each row, the sign rule gives $p_i(a_i^\top x - b_i) \geq 0$. For each variable, the coefficient rule gives $(c_j - p^\top A_j)x_j \geq 0$.

$$c^\top x - p^\top b = \sum_j (c_j - p^\top A_j)x_j + \sum_i p_i(a_i^\top x - b_i).$$

Every term is nonnegative. Hence the whole difference is nonnegative, which is weak duality.

Equality rows contribute zero. Unrestricted primal variables have zero coefficient differences. The remaining cases follow their stated signs.

Why weak duality rules out the unbounded–feasible combinations

Suppose the dual has a feasible p . Its finite value $b^\top p$ is a lower bound on every primal feasible cost. The primal therefore cannot decrease without bound.

Suppose the primal has a feasible x . Its finite value $c^\top x$ is an upper bound on every dual feasible value. The dual therefore cannot increase without bound.

These are the contrapositives of the two unboundedness implications. Neither argument says that infeasibility forces the other side to be unbounded.